

Industrial Internship Report on

1. Prediction of Remaining Operational Cycles Before Failure for Turbofan Engine

2. Quality Prediction in a Mining Process

Prepared by

P Charmi Reddy

Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.

During the training, two Machine Learning problem statements were selected and studied: **Prediction of Remaining Operational Cycles Before Failure for a Turbofan Engine** and **Quality Prediction in a Mining Process**.

The first project focuses on predictive maintenance. It uses historical sensor and operational data from turbofan engines to estimate the number of operational cycles remaining before failure. The objective is to support early identification of equipment degradation and help reduce unexpected failures and maintenance-related downtime.

The second project focuses on industrial process quality prediction. It uses process measurements from an iron ore flotation plant to predict the percentage of silica present in the final iron ore concentrate. Accurate prediction can help engineers identify potential quality issues earlier and take corrective actions.

This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1 Preface

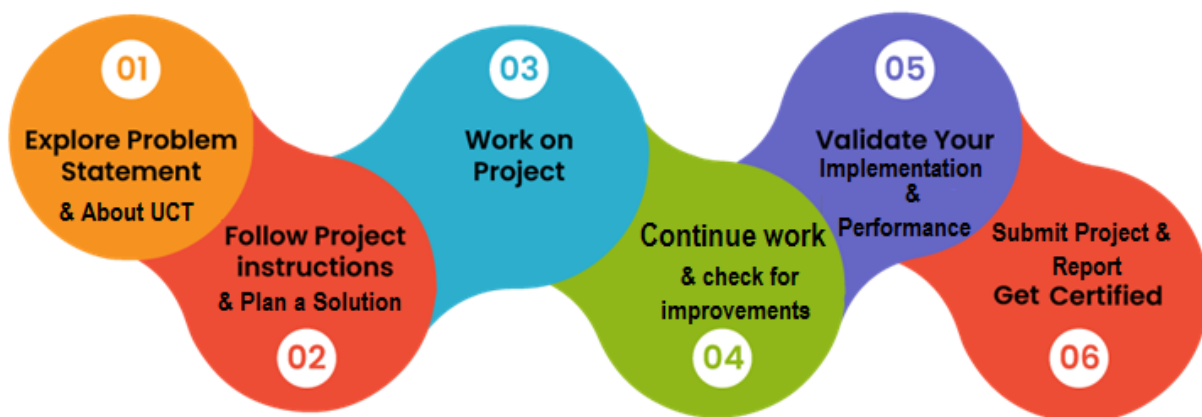
During the internship, I worked on two industrial Machine Learning problem statements. The first involved predicting the Remaining Useful Life of turbofan engines using sensor data. The second involved predicting silica concentration in an iron ore mining process.

The training was structured around understanding the problem, studying the dataset, preprocessing the data, performing exploratory analysis, developing Machine Learning models, evaluating their performance, and documenting the results.

The internship helped me improve my technical knowledge as well as my problem-solving and analytical skills. It also provided an understanding of how Machine Learning workflows differ depending on the characteristics of the industrial problem.

I would like to express my gratitude to the internship coordinators, mentors, faculty members, and everyone involved in providing guidance and support throughout the training.

I would also recommend similar industry-oriented projects to students because they provide an opportunity to connect academic concepts with practical applications.



2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



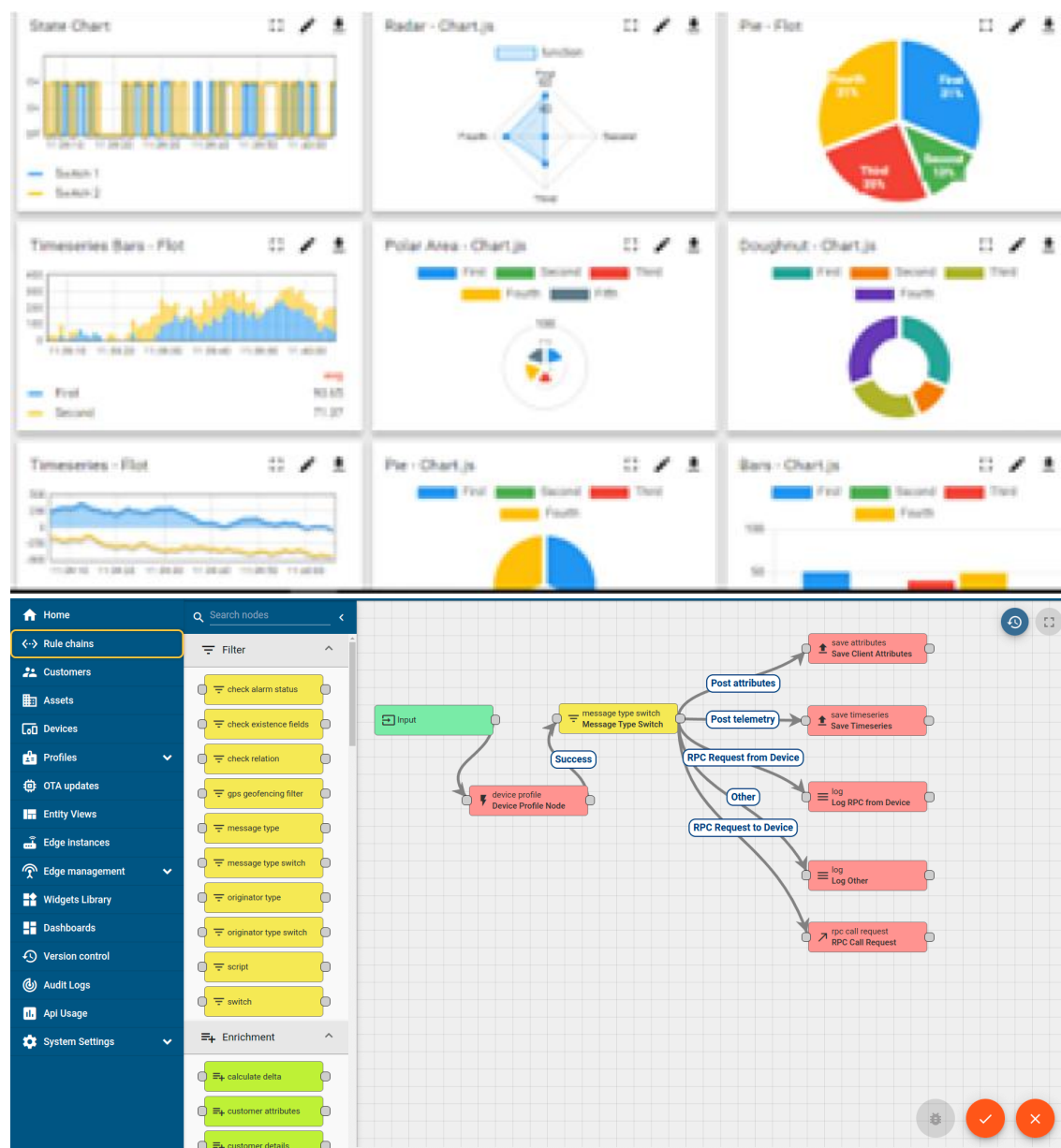
i. UCT IoT Platform (uct Insight)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY WATCH

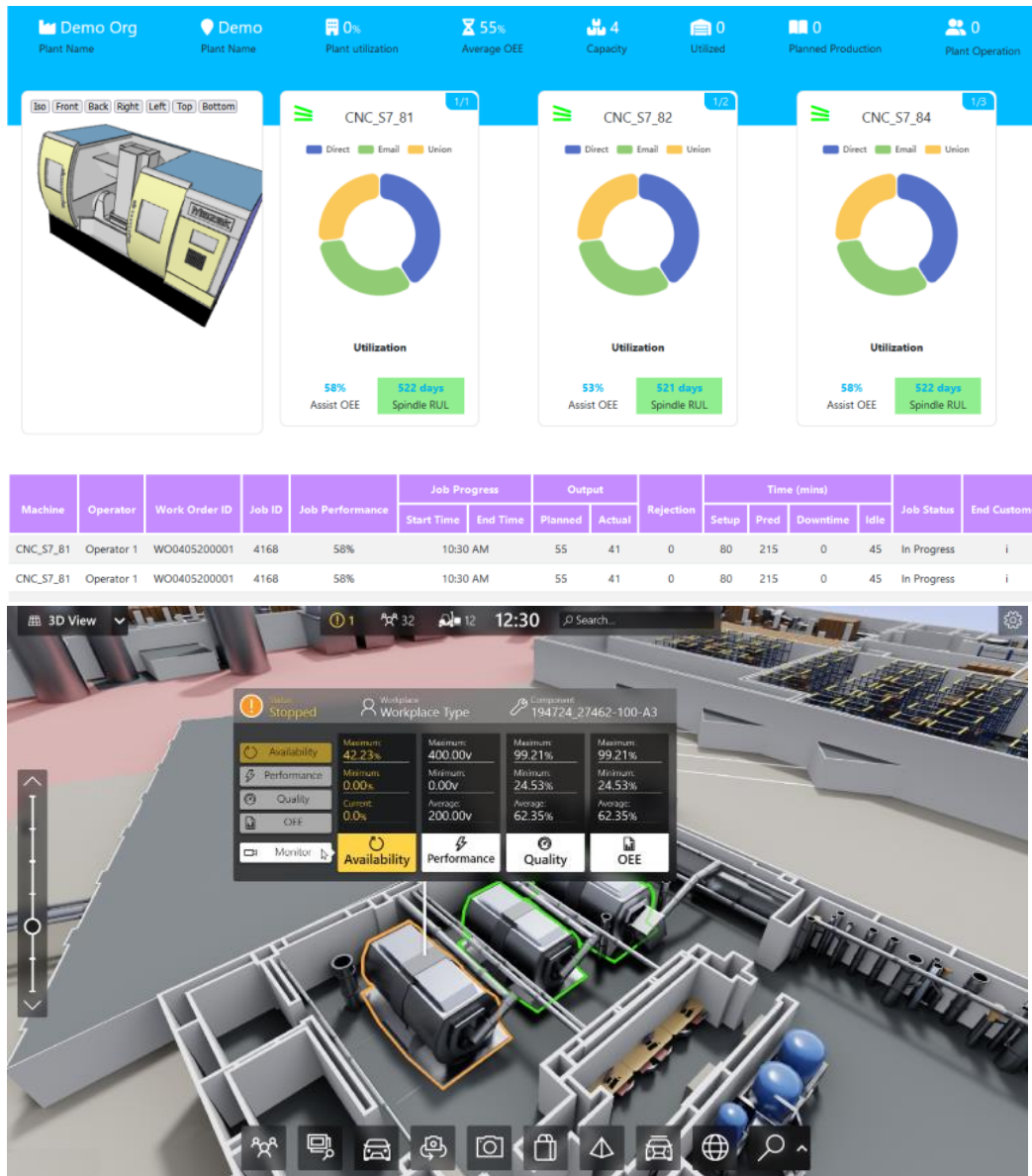
ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



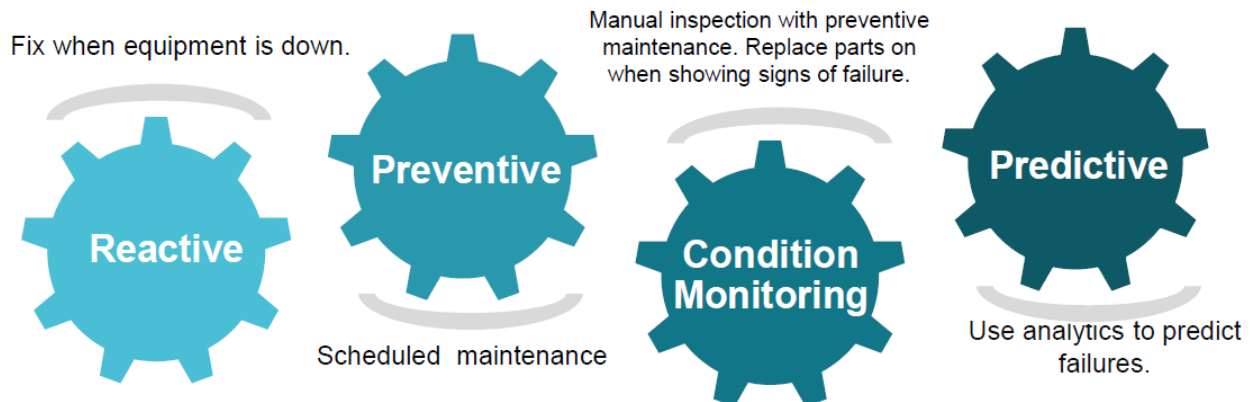


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

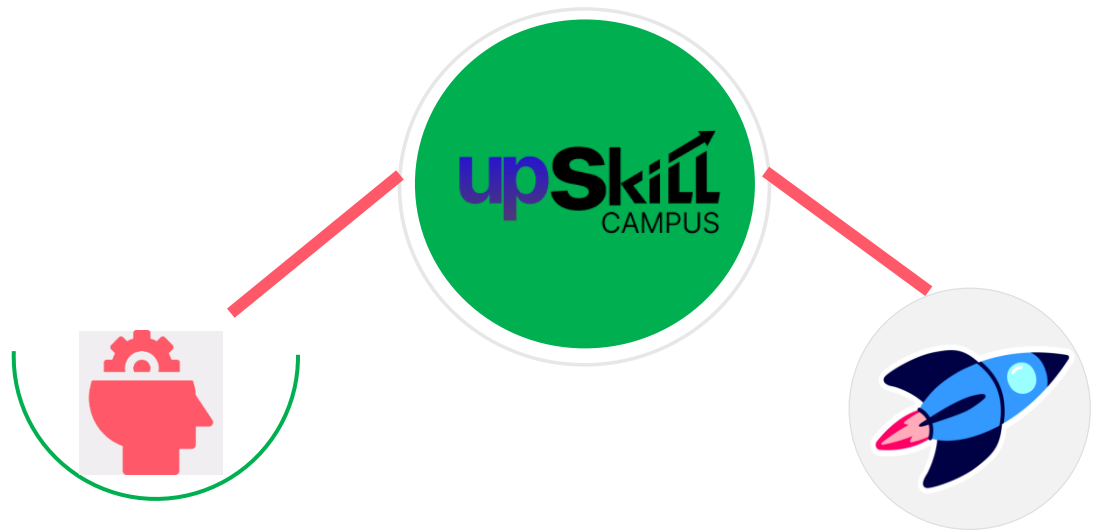
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

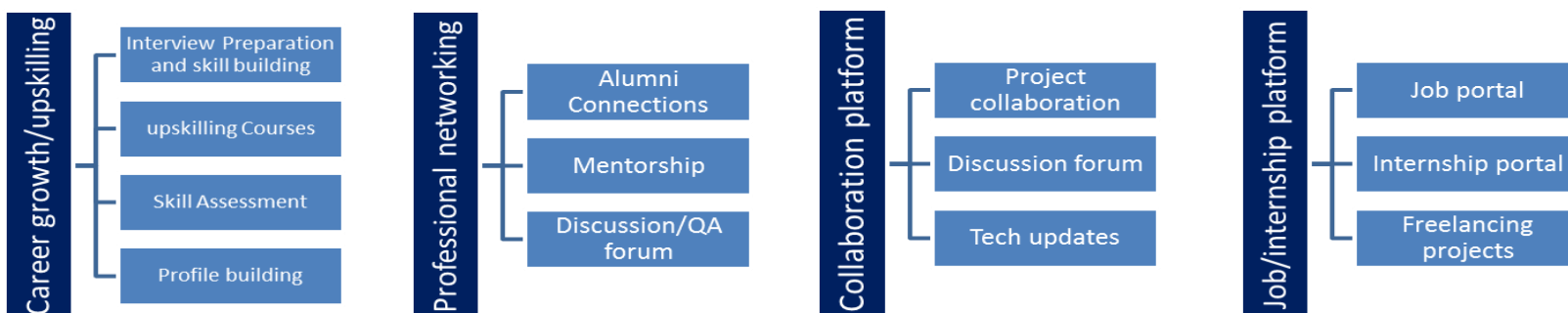
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

2.5 Reference

[1] The project datasets and descriptions provided as part of the internship were used as the primary references for the two projects.

[2] Additional technical references were used to understand Python, Machine Learning, data preprocessing, regression, predictive maintenance, and model evaluation concepts.

2.6 Glossary

Terms	Acronym
ML	Machine Learning
IoT	Internet of Things
EDA	Exploratory Data Analysis
OEE	Overall Equipment Effectiveness
MAE	Mean Absolute Error

3 Problem Statement

Project 1: Turbofan Engine Remaining Useful Life Prediction

Industrial equipment can gradually degrade during operation. If degradation is not identified in time, equipment failure can lead to unexpected downtime, maintenance costs, and production losses.

The project uses multivariate time-series data collected from turbofan engines. Each engine is represented by a sequence of operational cycles containing operational settings and sensor measurements.

The objective is to predict the **Remaining Useful Life (RUL)** of an engine, which represents the number of operational cycles remaining before failure.

The dataset contains multiple engine trajectories, operational settings, and sensor measurements. Different subsets of the dataset represent different operating conditions and fault modes.

Aim

To develop a Machine Learning-based approach that can estimate the remaining operational cycles of a turbofan engine before failure.

Project 2: Quality Prediction in a Mining Process

The second project focuses on an industrial flotation process used in an iron ore mining plant.

The quality of the final iron ore concentrate is affected by multiple process variables. One important quality measurement is the percentage of silica present in the concentrate.

The objective is to use process data to predict the silica concentration before the final laboratory measurement is available.

Aim

To develop a Machine Learning-based prediction system for estimating the percentage of silica in iron ore concentrate using process measurements.

4 Existing and Proposed solution

4.1 Existing Solution

Turbofan Engine

Traditional maintenance approaches may depend heavily on fixed maintenance schedules or responding after equipment faults occur.

These approaches can have limitations:

- Maintenance may be performed earlier than necessary.
- Unexpected failures may still occur.
- Sensor information may not be fully utilised.
- Maintenance planning can become less efficient.

Mining Process

In the mining process, quality measurements such as silica concentration may depend on laboratory analysis.

The limitation is that laboratory measurements may not always provide immediate information about changes in process quality.

This creates a need for predictive approaches that can provide earlier estimates.

4.2 Proposed Solution

The proposed approach uses Machine Learning to learn relationships between historical data and the desired prediction target.

Turbofan Engine

The proposed workflow is:

Engine Sensor Data → Data Preprocessing → Feature Preparation → Model Training → RUL Prediction → Model Evaluation

The model receives operational and sensor information and predicts the remaining operational cycles of the engine.

Mining Process

The proposed workflow is:

Process Data → Data Cleaning → Exploratory Analysis → Feature Preparation → Regression Model → Silica Prediction → Model Evaluation

The model uses process measurements to estimate the percentage of silica in the concentrate.

4.3 Value Addition

The proposed Machine Learning approaches can provide the following value:

- Early prediction of potential equipment failure.
- Support for predictive maintenance planning.
- Better utilisation of sensor and process data.
- Earlier identification of potential quality issues.
- Reduction of dependence on purely reactive decisions.
- Data-driven support for industrial engineers.
- Potential reduction in downtime and operational losses.

4.6 Code submission (Github link): [Repository Link](#)

4.7 Report submission (Github link) : [Report Link](#)

5 Proposed Design/ Model

5.1 High Level Diagram (if applicable)

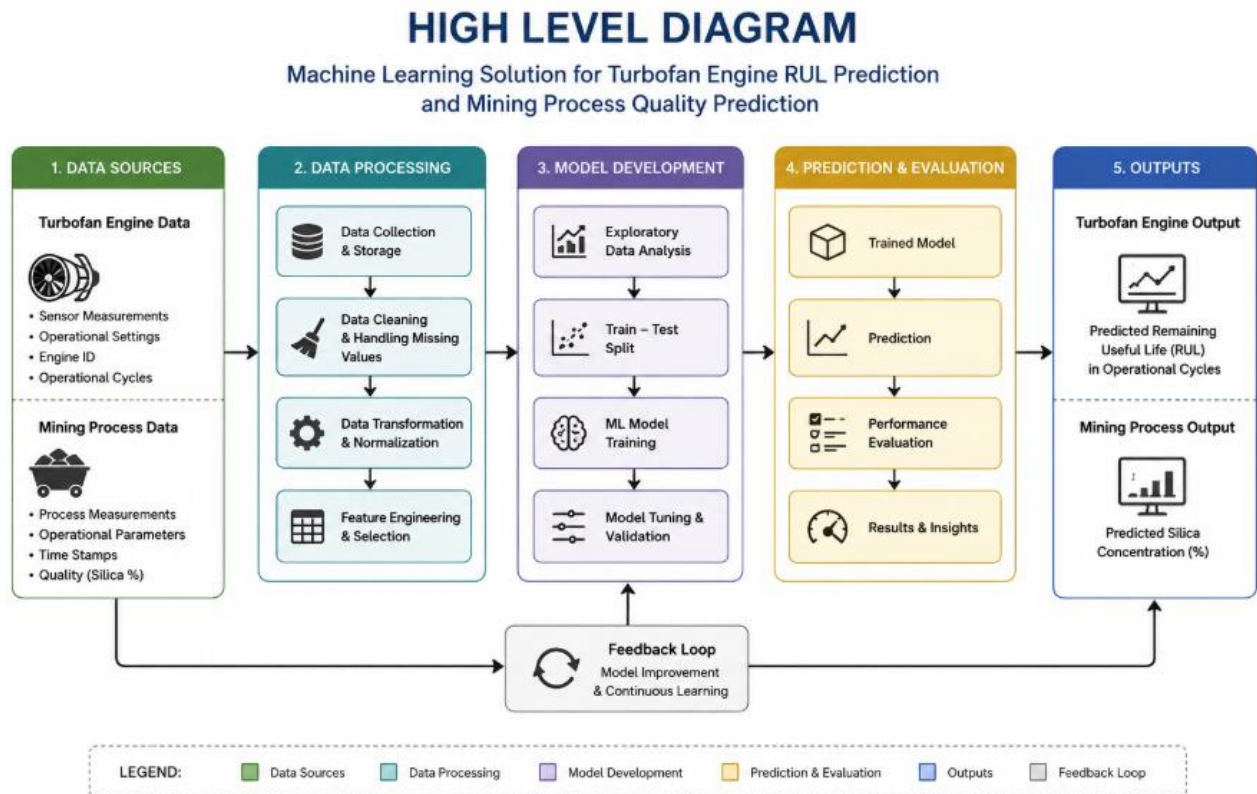
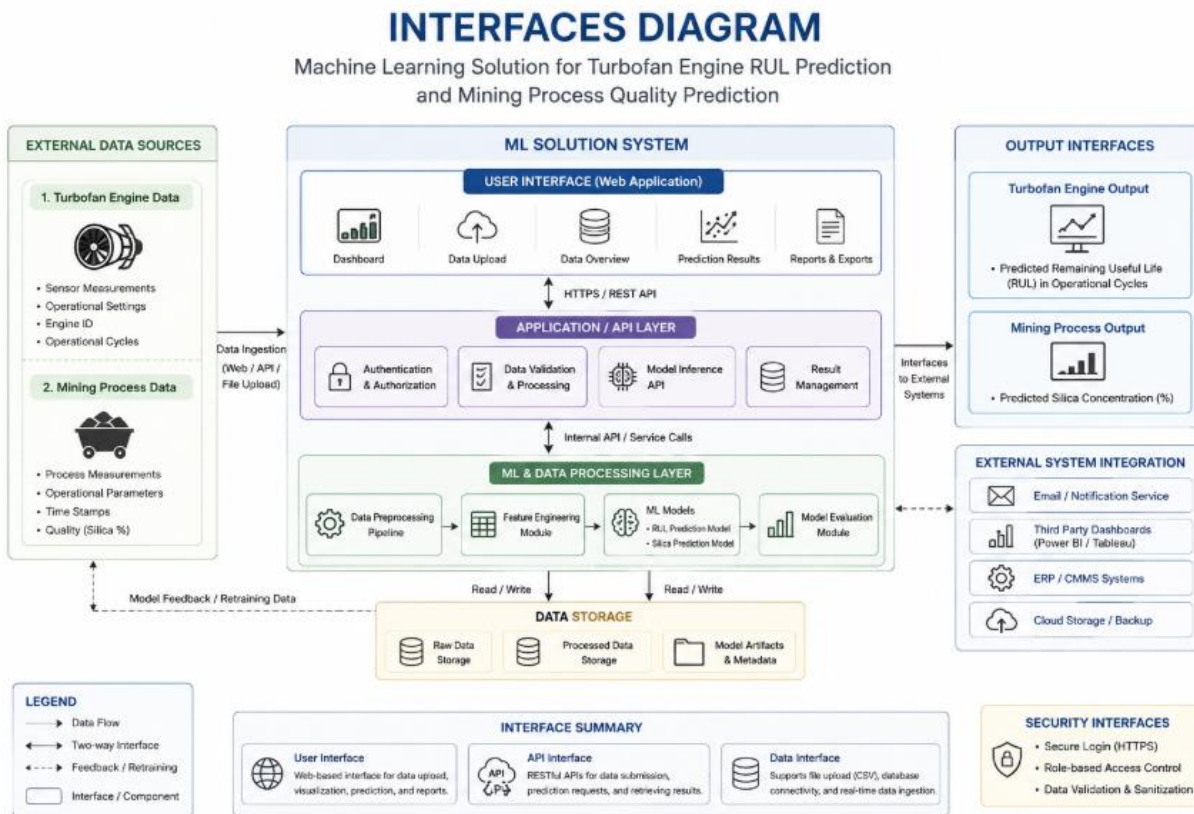


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Interfaces



6 Performance Test

This is very important part and defines why this work is meant of Real industries, instead of being just academic project.

Here we need to first find the constraints.

How those constraints were taken care in your design?

What were test results around those constraints?

Constraints can be e.g. memory, MIPS (speed, operations per second), accuracy, durability, power consumption etc.

In case you could not test them, but still you should mention how identified constraints can impact your design, and what are recommendations to handle them.

6.1 Test Plan/ Test Cases

Test Case 1: Dataset Validation

Objective: Verify that the datasets are loaded correctly.

Procedure:

- Load the datasets.
- Check the number of rows and columns.
- Inspect data types.
- Check missing values and invalid entries.

Expected Outcome:

The dataset should be correctly loaded and suitable for preprocessing.

Test Case 2: Preprocessing Validation

Objective: Verify that preprocessing produces consistent model-ready data.

Procedure:

- Apply the preprocessing steps.
- Check the resulting feature structure.
- Verify that required target variables are available.

Expected Outcome:

The processed data should be suitable for model training.

Test Case 3: Model Training

Objective: Verify that the Machine Learning models can be trained successfully.

Procedure:

- Split the prepared data appropriately.
- Train the selected model.
- Monitor the training process.

Expected Outcome:

The model should train successfully without data or implementation errors.

Test Case 4: Prediction

Objective: Verify that the trained model produces predictions.

Procedure:

- Provide unseen/test observations to the trained model.
- Generate predictions.
- Compare predictions with the expected target values.

Expected Outcome:

The model should produce valid numerical predictions.

Test Case 5: Model Evaluation

Objective: Measure prediction performance.

Procedure:

- Calculate appropriate evaluation metrics.
- Analyse prediction errors.
- Compare predicted and actual values.

Expected Outcome:

The evaluation should provide a measurable indication of model performance.

6.2 Test Procedure

The general testing procedure followed for the projects was:

1. Load the dataset.
2. Validate the input data.
3. Perform preprocessing.
4. Prepare the model features and target.
5. Train the baseline model.
6. Generate predictions.
7. Evaluate the predictions.
8. Analyse errors.
9. Refine the model where required.
10. Compare the refined approach with the baseline.

6.3 Performance Outcome

The projects were evaluated based on their ability to predict their respective target variables.

For the turbofan project, the primary performance objective is accurate prediction of Remaining Useful Life.

For the mining project, the primary performance objective is accurate prediction of silica concentration.

The final numerical performance values, including evaluation metrics and model comparisons, should be added after completing the implementation and testing of the models.

The performance analysis will also consider the trade-off between model complexity, computational requirements, prediction quality, and practical industrial applicability.

7 My learnings

The internship provided practical exposure to the complete Machine Learning development process.

Technical Learnings

- Improved understanding of Python for Machine Learning.
- Gained practical experience in dataset loading and preprocessing.
- Improved knowledge of Exploratory Data Analysis.
- Learned how to analyse industrial sensor data.
- Learned about time-series data and operational-cycle-based analysis.
- Understood the concept of Remaining Useful Life prediction.
- Gained exposure to regression-based quality prediction.
- Learned the importance of feature selection.
- Improved understanding of model evaluation and error analysis.
- Learned the importance of comparing baseline and refined models.

Problem-Solving Learnings

Working with two different industrial problems showed that Machine Learning cannot be approached using a single fixed workflow.

The turbofan project required understanding degradation patterns and operational cycles, while the mining project required understanding process variables and their relationship with quality measurements.

This helped me understand the importance of first studying the problem and dataset before selecting a modelling approach.

Career Development

The internship improved my confidence in working with real-world datasets and applying Machine Learning concepts beyond academic examples.

It also helped me understand how data scientists approach industrial problems, starting from a problem statement and progressing towards a measurable prediction system.

8. Future work scope

Several improvements can be explored beyond the scope of the internship.

Turbofan Engine Project

Future work can include:

- Testing additional Machine Learning algorithms.
- Exploring advanced time-series models.
- Improving feature engineering.
- Incorporating more advanced degradation indicators.
- Testing the approach across different operating conditions.
- Developing an automated predictive maintenance dashboard.
- Integrating real-time sensor data for continuous RUL prediction.
- Investigating model explainability for maintenance engineers.

Mining Process Project

Future work can include:

- Testing additional regression algorithms.
- Improving time-based feature engineering.
- Investigating short-term and long-term silica forecasting.
- Developing real-time prediction capabilities.
- Integrating the model with industrial monitoring systems.
- Creating alerts when predicted quality exceeds an acceptable range.
- Exploring optimisation techniques to recommend process adjustments.
- Investigating whether the model can operate without highly correlated quality variables.

Overall Future Scope

Both projects can potentially be extended from offline Machine Learning experiments into real-time industrial systems.

A complete industrial deployment could combine:

Sensors → IoT Platform → Data Processing → Machine Learning Model → Prediction → Dashboard / Alert → Engineer Action

Such a system could provide continuous data-driven support for predictive maintenance and industrial process optimisation.